TAKAHIRO MAEDA

Machine Learning Researcher at Toshiba Corporation
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HIGHLIGHTS

- Strong publications including first-authored CVPR, ICCV, RA-L papers.
- Expertise in enterprise-tuned VLMs, flow models, and safety-critical prediction systems.
- Worked at Toyota and CMU. Won 1st place in MPII Real Robot Challenge (Phases 2 & 3).

WORK EXPERIENCE

Toshiba Corporation, Japan — Corporate Laboratory *April 2025 - Present*
Research Engineer

- Prototyping domain-adapted VLMs for enterprise document understanding, focusing on model robustness and handling domain-specific terminology.
- Developing a context-engineering framework for AI agents to more reliably extract domain knowledge and perform tool-using tasks.
- Created an initial internal benchmark for technical drawing interpretation to standardize evaluation of multimodal reasoning models.
- Collaborating with business units to identify workflow automation opportunities and define requirements for future AI-agent integrations.

Toyota Technological Institute, Japan *April 2021 - September 2021*
Researcher

- Conducted research on computer vision and human motion prediction, continued in the subsequent doctoral program.

EDUCATION

Toyota Technological Institute, Japan *November 2021 - March 2025*
Doctoral Program, Electronics and Information Engineering (completed course requirements; withdrew without degree)

Toyota Technological Institute, Japan *April 2019 - March 2021*
Master of Engineering, Electronics and Information Engineering

Toyota Technological Institute, Japan *April 2015 - March 2019*
Bachelor of Engineering

RESEARCH INTEREST

Computer vision, robotics, human motion prediction, and multimodal models. Currently developing enterprise VLMs and AI agents at Toshiba, with long-term interests in robust perception for intelligent assistive robots.

PUBLICATIONS

T. Maeda, J. Cao, N. Ukita, and K. Kitani.
CacheFlow: Fast Human Motion Prediction by Cached Normalizing Flow *TMLR, 2026*

H. Taketsugu, T. Oba, **T. Maeda**, S. Nobuhara, and N. Ukita.
Physical plausibility-aware trajectory prediction via locomotion embodiment *CVPR, 2025*

T. Maeda, K. Takeshita, N. Ukita, and K. Tanaka.
Multimodal Active Measurement for Human Mesh Recovery in Close Proximity *RA-L, 2024*

T. Maeda and N. Ukita.
Fast Inference and Update of Probabilistic Density Estimation on Trajectory Prediction *ICCV, 2023*

T. Maeda and N. Ukita.
MotionAug: Augmentation with Physical Correction for Human Motion Prediction. *CVPR, 2022*

T. Yoneda, C.Schaff, **T. Maeda**, & M. Walter.
Grasp and motion planning for dexterous manipulation for the real robot challenge. *ArXiv, 2021*

T. Matsumoto, K. Shimosato, **T. Maeda**, T. Murakami, K. Murakoso, K. Mino, & N. Ukita.
Automatic Human Pose Annotation For Loose-Fitting Clothes. *MVA, 2019*

SKILLS

LLM/VLM Training	Domain adaptation (SFT, RLVR, OPSD)
LLM/VLM Evaluation & Ops	Benchmark construction, prompt/context optimization Inference and deployment with vLLM, Ollama, Hugging Face
AI Agents	ReAct, Tool Use, Computer Use; LangChain, Google ADK
AI Coding	Claude Code, Codex
Generative Models	Normalizing Flow, Diffusion, Flow Matching
Robotics	ROS, Toyota Human Support Robot
English	TOEFL iBT 94 (R:29 L:24 S:18 W:23, July 2024)

LICENSES

Certified Care Worker（介護福祉士国家資格） April 2022, Japan

EXPERIENCES

Robotics Institute, Carnegie Mellon University — Research Internship *Sep-Dec 2024*

Frontier Research Center, Toyota Motor Corporation — Research Internship *May-Aug 2023*

Toyota Technological Institute at Chicago — Research Internship *Sep-Dec 2019*

SERVICES

Program Committee member for AAAI 2027.
Conference reviewer for CVPR, ICCV, NeurIPS, and ICRA.
Journal reviewer for RA-L, TPAMI, and CVIU.

HONORS & AWARDS

Conference Presentation Award
IEEE Japan Nagoya Section, 2023

1st Place (Phase 2, €3500) & **1st Place** (Phase 3, €9000)
Real Robot Challenge 2021, the Max Planck Institute for Intelligent Systems

Best Poster Award
Machine Vision and Applications Organization, Tokyo, 2019